CHAPTER 4

Collection of Biological Materials

for Wildlife Disease Diagnosis

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1. Why We Collect Biological Materials — The Diagnostic Pipeline

Wildlife disease diagnosis works through several layers of investigation, each requiring specific biological materials collected at the right time, with the right preservation method, and dispatched to the appropriate laboratory. Understanding this diagnostic pipeline is essential before selecting which samples to collect.

The diagnostic pipeline encompasses:

- **Clinical observations** of the live or recently dead animal.
- **General health parameters** — invasive (haemato-biochemical profile) and non-invasive (scat/dung analysis).
- **Collection of specific biological materials** from diseased or dead animals.
- **Post-mortem (necropsy) examination** — gross lesions, histopathology (including histochemistry and immuno-histochemistry), microbiology (microscopy, culture, biochemical reactions, serological ID, molecular biology / PCR), toxicology, and parasitology.

How Disease Shows Up: Gene → Gross

Lesions develop in a sequence from the molecular level outward. Knowing this hierarchy tells you which diagnostic test detects what — and therefore which sample to collect and preserve.

Level	What Changes	How It Is Detected
1. Gene / Molecular	DNA mutation, epigenetic & gene-expression changes, abnormal protein synthesis	Not visible by light microscopy — requires PCR / molecular methods
2. Ultrastructural	Mitochondrial swelling, RER dilatation, ribosome detachment, membrane blebbing, chromatin clumping	Electron microscopy only
3. Light-Microscopic	Cellular swelling, fatty change, necrosis, apoptosis, inflammation, fibrosis, hyperplasia / dysplasia / neoplasia	Routine histopathology (H&E; stain)
4. Gross (Macroscopic)	Organ enlargement, tumour mass, ulcer, infarct, haemorrhage, abscess, tissue necrosis	Visible to the naked eye at necropsy

Key Concept: Lesion Hierarchy

Molecular lesion → Ultrastructural lesion → Microscopic lesion → Gross lesion

Most field investigations start at the gross level — but PCR can detect molecular lesions before any gross changes appear. Collect samples for BOTH.

2. Clinical Examination & the Samples It Generates

Common clinical workups include physical examination, haematology, serum biochemistry, urine examination, faecal examination, biopsy/FNAC, lateral flow assay, X-ray, and ultrasonography. For each workup, specific biological materials must be collected correctly to obtain valid results.

Clinical samples commonly collected from elephants:

- **Blood in anticoagulant vials** — EDTA *or* heparin (4–5 mL)
- **Serum** in a plain tube (allow to clot, then separate)
- **Faecal / rectal swabs or dung**
- **Ocular / nasal / oral swabs** — with or without Viral Transport Medium (VTM)
- **Trunk wash fluids** — critical for TB and EEHV diagnosis in elephants
- **Skin scrapings / hair follicles**
- **Any lesions** — abscess / wound discharge / tissue biopsy

Planning Rule

Always plan around two questions before collecting any sample:
(1) Which laboratory test is intended for this sample?
(2) What biosafety precautions and preservation method are required?

3. Master Sample-Collection Table

For each sample type: why you collect it, how to preserve / collect it, and how to store and transport it to the laboratory.

Sample	Purpose	Preservative / Collection Method	Storage & Transport
Blood	Infections, anaemia, general health	5–6 mL in EDTA vial	4–8 °C (refrigerated)
Blood smear	Blood parasites, cell abnormalities	Methanol-fixed blood smear on clean glass slide	Wrap in dry soft paper; room temperature
Plasma	Toxicity & drug-metabolite analysis	5–6 mL in heparin vial; transfer plasma to sterile tubes	–20 °C / –80 °C, or dry ice
Serum	Infections (antigen & antibody), health parameters, hormones	5–6 mL in plain serum vial; after clotting transfer serum to sterile tubes	4–8 °C short term; –20/–80 °C long term; dry ice for transport
Dung (faeces)	Parasites	20 g in sterile vial / zip-lock; worms in 10% formalin or 70% ethanol	4–8 °C; room temperature
Dung	Digestive disorders	100–200 g in sterile vial	4–8 °C
Dung	Hormone analysis	100–200 g in sterile vial / zip-lock	4–8 °C
Urine	Kidney function, UTIs	20–30 mL aseptically in sterile vial	4–8 °C
Urine	Bacterial infections (e.g. Leptospira)	Drops into a tube with specific bacterial media (e.g. Leptospira media)	4–8 °C
Saliva	Bacterial & viral infections	1–2 mL in sterile tubes or sterile swabs	4–8 °C (bacteria); –20/–80 °C (viruses)

Sample	Purpose	Preservative / Collection Method	Storage & Transport
Nasal discharge / trunk wash	Respiratory disease, esp. TB & EEHV	20–30 mL trunk wash using normal saline in sterile vial	–20/–80 °C for EEHV & TB; 4–8 °C for bacteria
Ocular discharge	Eye infections	Sterile tube/swab, ± specific media	4–8 °C (bacteria); –20/–80 °C (viruses)
Biopsy tissue / skin swabs	Skin & tissue diseases	Swabs aseptically ± transport media; tissues in 10% buffered formalin	4–8 °C for swabs; room temperature for formalin-fixed tissue

4. Examination-Specific Protocols

4a. Histopathological Examination

- Collect tissue **soon after death**; fix **immediately** in **10% buffered formalin**.
- Take several pieces — from the **centre** and from the **edge of the lesion** at the junction with healthy tissue (especially critical for large lesions).
- Tissue thickness: **0.5–1 cm**; formalin volume must be at least **10–15× the volume of the tissue** (minimum ratio: 10× the tissue amount).
- Use a **sharp instrument** (scalpel, knife, or razor) for a clean, crush-free cut.
- Always include **lesion tissue plus adjacent healthy tissue** in the same block.

Preparing 10% Neutral Buffered Formalin (NBF)

Commercial (saturated) formalin ≈ 37–40% formaldehyde. **10% NBF ≈ 4% formaldehyde** — this is the standard histopathology fixative. Tissue : fixative ratio = **1 : 10** at minimum (e.g. 10 g tissue in 100 mL of 10% NBF).

Ingredient	Quantity
Saturated formalin (37–40% formaldehyde)	100 mL
Distilled water	900 mL
Sodium phosphate monobasic (NaH■PO■·H■O)	4.0 g
Sodium phosphate dibasic (Na■HPO■)	6.5 g
TOTAL	1,000 mL (1 litre)

Method: Dissolve both phosphate salts fully in ~800 mL distilled water → add 100 mL saturated formalin → top up to 1,000 mL → mix → label (name, concentration, date, preparer).

Simpler unbuffered version: 100 mL saturated formalin + 900 mL distilled water (≈ 4% formaldehyde) — adequate when buffered NBF is unavailable in the field.

Preparation Notes

Use clean glass or plastic containers.
Dissolve phosphate salts BEFORE adding formalin.
Store at room temperature, tightly capped; label with name, concentration, and date.

4b. Bacteriological & Virological Examination

- Keep everything **free from contamination**; use **sterile instruments** (scalpels, scissors, syringes, bottles, slides, swabs, polythene bags).
- **Disinfect skin** before the primary incision to prevent surface contamination.
- Sear the organ surface (e.g. heart) with a **hot iron spatula** before sampling heart blood — to sterilise the surface and eliminate external contamination.
- Sample **abnormal areas** near the **edge of affected tissue** where live organisms are most likely still viable.
- **Appropriate samples:** whole blood, serum, CSF, pus, abscess/nodule areas, intestinal contents (within a tied loop), smears of pus or infected tissue.

For virology: use 5–10× volume of sterile 50% buffered glycerol, chilled PBS (pH 7.4), or Viral Transport Medium (VTM). Other acceptable preservatives: RNA Later, 70% ethanol (for PCR-based detection).

Transport: bacteria at 4 °C; viruses at 4 °C or on dry ice / frozen.

Heart-blood smears: heat-fix for bacteria only; fix in absolute methanol for haemoparasites.

4c. Parasitological Examination

- **Blood smears** on clean glass slides — fix in **absolute methanol**.
- **Skin scrapings** → preserve in **10% KOH**.
- **2 g faeces** → **70% ethyl alcohol** for molecular work; **formalin** for morphology-based identification.
- **Larvae, worms, maggots** → **70% alcohol** or **10% formalin** (room temperature).
- **Protozoan diseases** → lymph nodes, blood, and faecal sample in **5% formalin**.

4d. Toxicological Examination

- Collect ≥ **100 g of stomach + intestinal contents** and ≥ **100 g of liver + kidney** in a glass container with saturated salt solution* or ice.
- Also collect **environmental samples** — baits, feed/fodder, water sources near the mortality site.
- **Pack each sample type separately** in an ice box; do not mix organ samples with GI contents or environmental samples.
- All containers must be **serially numbered, properly labelled, and kept under safe custody** — chain of custody is legally critical in forensic cases.

■ Important — Qualitative vs. Quantitative Toxicology

*Saturated salt solution is NOT used when toxicants must be quantified.
For quantitative toxicology: use sterile containers / zip-locks and deep-freeze at –20 °C.

Toxicology sample guide — ante-mortem, post-mortem, and environmental:

Sample	Amount	Notes
ANTE-MORTEM		
Whole blood	5–10 mL	EDTA or heparin vial
Serum	5–10 mL	Remove from clot; use trace-element tubes for zinc assays
Urine	25–50 mL	Plastic screw-capped tube
GI contents	≥ 100 g each	Representative stomach/rumen contents; faeces
Hair	1–2 g	Clean zip-lock pouch
Milk	30 mL	Sterile collection tube

Sample	Amount	Notes
CSF	1–2 mL	Sterile tube; refrigerate immediately
POST-MORTEM		
Blood/serum/urine (from heart)	As ante-mortem	As per ante-mortem guidelines above
Liver	100–250 g	Plastic container; –20 °C
Kidney	100–250 g	Plastic container; –20 °C
Brain	One half	Sagittal cut; plastic; –20 °C
Fat	100 g	Foil inside plastic bag; –20 °C
GI contents	≥ 100 g each	Representative samples; PACKAGE SEPARATELY from organs
Ocular fluid / eyeball	As much as possible	Aqueous fluid preferred; or whole eyeball
Bone	≥ 100 g	For heavy metal analysis
Spleen	100 g	As for liver
Lung	100 g	As for liver
Hair or skin	—	If dermal exposure suspected
ENVIRONMENTAL		
Bait / source material	200 mL or g	Clean jar / whirl-pak; label precisely
Feed	≥ 500 g	Composite sample per lot/batch
Plants	Whole or representative	Fresh-pressed-dried, or frozen
Water	≥ 1 L	Clean glass jar; plastic lid for metals analysis

4e. Pregnancy Diagnosis & Deficiency Diseases

Pregnancy diagnosis in carnivores and elephants: collect **100–150 g of dung** plus serum samples for hormone analysis.

Deficiency diseases: collect **serum samples** for macro- and micro-nutrient assessment.

5. Elephant-Specific Collection Notes

- **Blood:** drawn from an **ear vein**; make blood smears immediately and fix in 70% / absolute methanol.
- **Trunk wash fluid:** the elephant draws water into the trunk, washes the inside, and the used water is expelled into a clean container. This is the critical sample for **TB and EEHV** diagnosis and should be stored at –20/–80 °C.
- **Rectal and oral swabs:** collected with sterile swab sticks, placed in zip-lock pouches (± VTM); store in RNA storage tubes or RNA Later as needed for molecular diagnosis of EEHV.

Elephant Priority Sample: Trunk Wash

Trunk wash is the MOST IMPORTANT sample for EEHV and TB diagnosis in elephants.
Freeze at –80 °C (or on dry ice for transport) if EEHV is suspected.
Do not refrigerate only — EEHV degrades rapidly at 4 °C.

6. Packing & Transport

Preferred Method — Fresh & Frozen Tissue

a. Place **absorbent material** (cat litter or paper towel) in the outer bag to absorb spills.

b. Tissues → individual, sterile, **labelled** plastic bottles or whirl-type bags.

c. Containers + **frozen refrigerant packs** → heavy plastic bag labelled with contents.

d. Bag → **insulated cooler** for shipment.

e. Specimen submission sheet + all documents → envelope → plastic bag → taped to the **outside** of the package.

10-Step Packing Protocol (Zoonotic / General Samples)

1. Collect preferred samples with **all biosafety precautions** (gloves, mask, PPE).

2. Label specimens (animal name/species, age, gender, specimen ID, date, site).

3. Parafilm-seal the neck of all vials and containers.

4. Cover vials with **absorbent material (cotton)**.

5. Put tubes/vials in a **zip-lock pouch** and seal.

6. Place that pouch inside a **plastic container or another zip-lock** and seal tightly.

7. Put the container in a **thermocol box surrounded by hard-frozen gel packs**.

8. Put the **history form** in a separate zip-lock pouch inside the thermocol box.

9. Close and seal the thermocol box with adhesive tape.

10. Write the **address of the testing laboratory** clearly on top of the box.

Do's and Don'ts

Do ✓	Don't ✗
Use clean, properly labelled, leak-proof containers	Submit organs wrapped in plain plastic wrap
Use adequate fixative (10× formalin for histopath)	Use dirty or leaking jars / bottles
Parafilm-seal vial necks before packing	Under-fix tissue (too little formalin or too thick)
Maintain cold chain — refrigerate bacteria; freeze viruses	Ship a whole head or carcass loose in a leaking, poorly packed box
Include a complete history form and transport certificate	Mix samples from different organ types in the same container
Serial-number and record all forensic/toxicology samples	Break chain of custody for forensic samples

Transport Certificate

Include a signed certificate with every shipment stating that the samples are **non-corrosive, non-inflammable, and non-infectious**. The certificate must name the sender, species, sample type, origin, destination laboratory, and preservation method (ice gel packs / dry ice).

Example Certificate Wording

Sample wording: "Certified that samples of an Asian elephant (blood and trunk secretions) from [site], [date], are being transported by [scientist name] to ICAR-IVRI, Bareilly, for health examination; securely packed in sterile containers with frozen gel packs."

7. Essentials for Biological Sampling — Quick Master Reference

Diagnostic Activity	Specimen	Preservative / Container	Key Notes
Haematology	Whole blood	EDTA/heparin vial; ice pack 4–8 °C	Gently rotate to mix; do not keep at room temp for long
Serum biochemistry	Serum	Serum tube ± clot activator/gel; 4–8 °C or freeze	Handle gently; AVOID haemolysis
Histopathology	Tissues — liver, lung, kidney, spleen, heart, stomach, intestine, brain, lymph node, tumours (normal + abnormal)	10% NBF; leak-proof container; room temperature	0.5–1 cm thick; formalin:tissue ratio ≥ 10:1; include lesion + healthy margin
Bacteriology	Organs, ocular/rectal/naso- & oro-pharyngeal swabs, heart blood, exudates, intestinal loops, urine	Refrigerate 4–8 °C; nutrient broth / peptone water / PBS	Avoid contamination; sample type varies by suspected disease
Virology	Organs (small pieces), swabs, exudates, trunk wash, body fluids	VTM or PBS (pH 7.4); 50% glycerol saline	Refrigerate or dry ice; varies by disease
Parasitology — endoparasites	Nematodes, trematodes, cestodes	70% alcohol or 5–10% formalin	Room temperature
Parasitology — ectoparasites	External parasites (ticks, mites, flies)	10% KOH or 70% alcohol	Room temperature
Parasitology — blood	Blood parasites	Methanol-fixed blood smear	Glass slide at room temperature
Toxicology	~100 g organs, GI contents, blood, fat, water, fodder, feed	Qualitative: saturated salt solution. Quantitative: sterile containers; deep freeze –20 °C	Refrigerate for qualitative; freeze for quantitative; separate containers
Pregnancy diagnosis	Serum, scat/dung (100–200 g)	Plastic container / zip-lock; 4–8 °C	Refrigerate
Genetic / species ID	Tissues, blood, hair follicles, dung, bone marrow	Blood in EDTA; hair in pouches; tissue in 70% ethanol or dry ice	Refrigerate blood; freeze tissue/marrow; hair at room temp in dry pouch

8. Where to Send Samples

Services Available at ICAR-IVRI, Bareilly

- PCR diagnosis of important infectious diseases (EEHV, TB, Anthrax, FMD, Rabies, etc.)
- Bacterial isolation & antibiotic sensitivity testing (ABST)
- Toxicological investigations (qualitative and quantitative)
- Parasitology — endo-, ecto-, and haemoprotozoan parasites
- Pregnancy diagnosis via dung samples
- Necropsy, histopathology, and clinical pathology (blood, urine)
- Serodiagnosis of TB, Leptospirosis, and other infections

ICAR-IVRI Contact

Contact: In-charge, Centre for Wildlife
ICAR-IVRI, Izatnagar, Bareilly — cwlincharge@gmail.com

National Referral Laboratories by Specialty

Specialty	Recommended Laboratory
General wildlife / referral	National Referral Centre on Wildlife Health / Centre for Wildlife, ICAR-IVRI, Izatnagar
Exotic viral diseases (e.g. Avian Influenza)	ICAR-NIHSAD, Bhopal
Foot and Mouth Disease (FMD)	ICAR-Directorate on FMD, Bhubaneswar; IVRI Bengaluru
Toxicology	CSIR-Indian Institute of Toxicology Research (IITR), Lucknow
Forensic & molecular identification	Wildlife Institute of India (WII), Dehradun; CSIR-CCMB, Hyderabad
Poisoning incl. NSAIDs & avian forensic cases	SACON, Coimbatore (southern campus of WII)
Regional diagnostic labs (DADF)	Bangalore (KVAFSU Hebbal), Jalandhar, Pune (Aundh), Guwahati (Khanapara, NE region)
State veterinary universities / specialized centres	TANUVAS Chennai; KVASU Pookode Wayanad; OUAT Bhubaneswar; School of Wildlife Forensic & Health, Jabalpur (NDVSU); ICAR-NIVEDI Bengaluru

Key Takeaways

Chapter Summary — Biological Sample Collection

1. Match the sample, preservative, and cold-chain to the intended test — the same organ goes into formalin for histopath, VTM for virology, and salt solution / freezer for toxicology.
2. Fix histology tissue IMMEDIATELY, thinly (0.5–1 cm), in 10× formalin volume, including both lesion and healthy margin.
3. Sterility and anti-contamination technique are critical for all microbiology samples.
4. Label, document, and cold-chain every sample; always include a transport certificate.
5. Toxicology and forensic samples require serial numbering and strict chain of custody.
6. Trunk wash is the priority sample for EEHV and TB in elephants — freeze immediately.